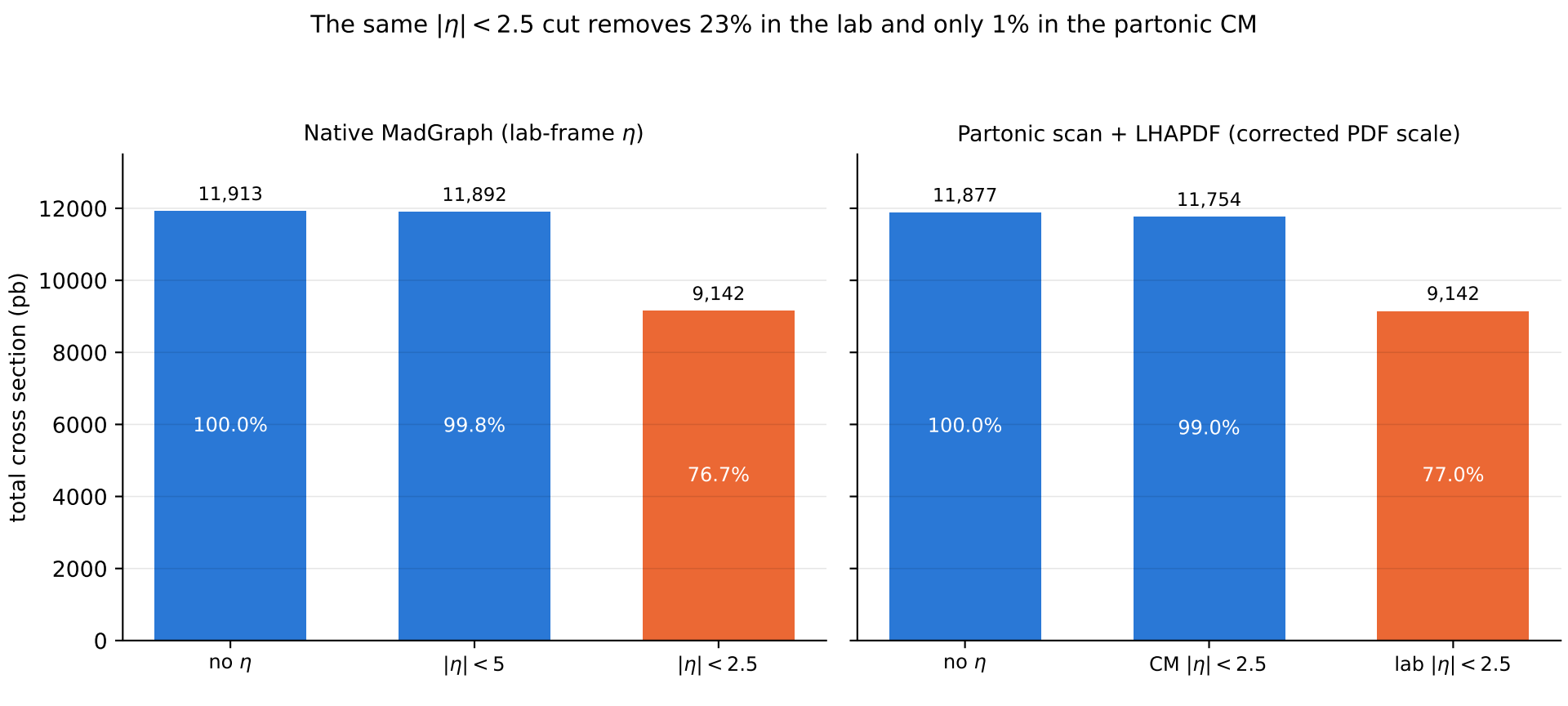
The same $|\eta| < 2.5$ cut removes 23% in the lab and only 1% in the partonic CM
Native MadGraph (lab-frame η)
Partonic scan + LHAPDF (corrected PDF scale)
total cross section (pb)
0
2000
4000
6000
8000
10000
12000
11,913
11,892
9,142
100.0%
99.8%
76.7%
11,877
11,754
9,142
100.0%
99.0%
77.0%
no η
$|\eta| < 5$
$|\eta| < 2.5$
no η
CM $|\eta| < 2.5$
lab $|\eta| < 2.5$